

Library Management System using Servlets and JSP

Objective

To develop a web-based Library Management System where:

- Librarians can add books, issue books, and track book availability.
- Users can search for books and view their borrowing history.

Technology Stack

- **Frontend:** JSP (JavaServer Pages), HTML
- **Backend:** Java Servlets
- **Server:** Apache Tomcat 9
- **IDE:** Eclipse IDE for Enterprise Java and Web Developers
- **Data Storage:** Session-based (No database used)

System Overview

- **Librarians** log in to manage books and issue them.
- **Users** log in to search available books and check their borrowing history.
- Role-based access is implemented using session attributes.
- Book data and borrowing records are stored in session objects.

Modules Description

Module	Files	Description
Login System	LoginServlet.java, login.jsp	Handles librarian and user login. Stores role in session.
Librarian Dashboard	librarianDashboard.jsp, BookServlet.java	Allows adding books and issuing books. Manages books in session.
User Dashboard	userDashboard.jsp, search.jsp, borrowHistory.jsp	Allows searching books and viewing borrowing history.
Book Management	Book.java, BookServlet.java	Handles book objects and stores them in session.
Borrowing Management	Borrow.java, BorrowServlet.java, borrowHistory.jsp	Manages borrowing books and storing history in session.
Role-Based Access	Handled in every JSP file	Ensures only the correct role can access pages.

Code

//login.jsp

```
<html><body>
<h2>Login</h2>
<form action="LoginServlet" method="post">
  Username: <input name="username"><br>
  Password: <input type="password" name="password"><br>
  <input type="submit" value="Login">
</form>
<c:if test="${param.error == 1}"><p style="color:red">Invalid credentials</p></c:if>
</body></html>
```

// LoginServlet.java

```
package servlet;

import java.io.IOException;
import javax.servlet.*;
import javax.servlet.http.*;

public class LoginServlet extends HttpServlet {
    protected void doPost(HttpServletRequest request, HttpServletResponse response) throws
    ServletException, IOException {
        String username = request.getParameter("username");
        String password = request.getParameter("password");

        HttpSession session = request.getSession();

        if(username.equals("librarian") && password.equals("lib123")) {
            session.setAttribute("role", "librarian");
            session.setAttribute("username", username);
            response.sendRedirect("librarianDashboard.jsp");
        } else if(username.equals("user") && password.equals("user123")) {
            session.setAttribute("role", "user");
            session.setAttribute("username", username);
            response.sendRedirect("userDashboard.jsp");
        } else {
            response.sendRedirect("login.jsp?error=1");
        }
    }
}
```

//BookServlet.java

```
package servlet;

import java.io.IOException;
import java.util.*;
import javax.servlet.*;
```

```

import javax.servlet.http.*;
import model.Book;

public class BookServlet extends HttpServlet {
    protected void doPost(HttpServletRequest request, HttpServletResponse response) throws
ServletException, IOException {
        HttpSession session = request.getSession();
        List<Book> books = (List<Book>) session.getAttribute("books");
        if (books == null) {
            books = new ArrayList<>();
        }

        String id = request.getParameter("id");
        String title = request.getParameter("title");
        books.add(new Book(id, title));
        session.setAttribute("books", books);
        response.sendRedirect("librarianDashboard.jsp");
    }

    protected void doGet(HttpServletRequest request, HttpServletResponse response) throws
ServletException, IOException {
        String keyword = request.getParameter("keyword");
        HttpSession session = request.getSession();
        List<Book> books = (List<Book>) session.getAttribute("books");
        List<Book> results = new ArrayList<>();
        if (books != null) {
            for (Book b : books) {
                if (b.getTitle().toLowerCase().contains(keyword.toLowerCase())) {
                    results.add(b);
                }
            }
        }
        session.setAttribute("searchResults", results);
        response.sendRedirect("search.jsp");
    }
}

```

//BorrowServlet.java

```

package servlet;

import java.io.IOException;
import java.util.*;
import javax.servlet.*;
import javax.servlet.http.*;
import model.*;

public class BorrowServlet extends HttpServlet {
    protected void doPost(HttpServletRequest request, HttpServletResponse response) throws
ServletException, IOException {
        HttpSession session = request.getSession();

```

```

        List<Book> books = (List<Book>) session.getAttribute("books");
        List<Borrow> borrows = (List<Borrow>) session.getAttribute("borrows");
        if (borrows == null) {
            borrows = new ArrayList<>();
        }

        String bookId = request.getParameter("bookId");
        String username = (String) session.getAttribute("username");

        for (Book b : books) {
            if (b.getId().equals(bookId) && b.isAvailable()) {
                b.setAvailable(false);
                borrows.add(new Borrow(username, bookId));
                break;
            }
        }
        session.setAttribute("borrows", borrows);
        response.sendRedirect("userDashboard.jsp");

import java.io.IOException;
import java.util.*;
import javax.servlet.*;
import javax.servlet.http.*;
import model.*;

public class BorrowServlet extends HttpServlet {
    protected void doPost(HttpServletRequest request, HttpServletResponse response) throws
        ServletException, IOException {
        HttpSession session = request.getSession();
        List<Book> books = (List<Book>) session.getAttribute("books");
        List<Borrow> borrows = (List<Borrow>) session.getAttribute("borrows");
        if (borrows == null) {
            borrows = new ArrayList<>();
        }

        String bookId = request.getParameter("bookId");
        String username = (String) session.getAttribute("username");

        for (Book b : books) {
            if (b.getId().equals(bookId) && b.isAvailable()) {
                b.setAvailable(false);
                borrows.add(new Borrow(username, bookId));
                break;
            }
        }
        session.setAttribute("borrows", borrows);
        response.sendRedirect("userDashboard.jsp");
    }
}

```

//Book.java

```
package model;

public class Book {
    private String id;
    private String title;
    private boolean isAvailable = true;

    public Book(String id, String title) {
        this.id = id;
        this.title = title;
    }

    public String getId() { return id; }
    public String getTitle() { return title; }
    public boolean isAvailable() { return isAvailable; }
    public void setAvailable(boolean available) { isAvailable = available; }
}
```

//Borrow.java

```
package model;

import java.util.Date;

public class Borrow {
    private String username;
    private String bookId;
    private Date borrowDate;

    public Borrow(String username, String bookId) {
        this.username = username;
        this.bookId = bookId;
        this.borrowDate = new Date();
    }

    public String getUsername() { return username; }
    public String getBookId() { return bookId; }
    public Date getBorrowDate() { return borrowDate; }
}
```

//search.jsp

```
<%@ page import="java.util.*,model.Book" %>
<%
String role = (String) session.getAttribute("role");
if(!"user".equals(role)) {
    response.sendRedirect("login.jsp");
}
```

```

        return;
    }
    %>

<html><body>
<h2>Search Results</h2>
<%
List<Book> results = (List<Book>) session.getAttribute("searchResults");
if (results != null) {
    for (Book b : results) {
        if (b.isAvailable()) {
            %>
<form action="BorrowServlet" method="post">
    <%= b.getId() %> - <%= b.getTitle() %>
    <input type="hidden" name="bookId" value="<%= b.getId() %>">
    <input type="submit" value="Borrow">
</form>
<%
        } else {
            out.println(b.getId() + " - " + b.getTitle() + " (Borrowed)<br>");
        }
    }
}
%>
<a href="userDashboard.jsp">Back</a>
</body></html>

```

//borrowHistory.jsp

```

<%@ page import="java.util.*,model.*" %>
<%
String role = (String) session.getAttribute("role");
if(!"user".equals(role)) {
    response.sendRedirect("login.jsp");
    return;
}
%>

<html><body>
<h2>Your Borrowing History</h2>
<%
List<Borrow> borrows = (List<Borrow>) session.getAttribute("borrows");
String username = (String) session.getAttribute("username");
if (borrows != null) {
    for (Borrow br : borrows) {
        if (br.getUsername().equals(username)) {
            out.println("Book ID: " + br.getBookId() + " Date: " + br.getBorrowDate() + "<br>");
        }
    }
}
%>

```

```
<a href="userDashboard.jsp">Back</a>
</body></html>
```

//userDashboard.jsp

```
<%
String role = (String) session.getAttribute("role");
if(!"user".equals(role)) { response.sendRedirect("login.jsp"); return; }
%>

<html><body>
<h2>User Dashboard</h2>
<form action="BookServlet" method="get">
  Search Book: <input name="keyword">
  <input type="submit" value="Search">
</form>
<a href="borrowHistory.jsp">View Borrow History</a>
<br><a href="login.jsp">Logout</a>
</body></html>
```

//librarianDashboard.jsp

```
<%@ page import="java.util.*,model.Book" %>
<%
String role = (String) session.getAttribute("role");
if(!"librarian".equals(role)) { response.sendRedirect("login.jsp"); return; }
%>

<html><body>
<h2>Librarian Dashboard</h2>
<form action="BookServlet" method="post">
  Book ID: <input name="id"> Title: <input name="title">
  <input type="submit" value="Add Book">
</form>
<h3>Book List</h3>
<%
List<Book> books = (List<Book>) session.getAttribute("books");
if (books != null) {
  for (Book b : books) {
    out.println(b.getId() + " - " + b.getTitle() + " - " + (b.isAvailable()?"Available":"Borrowed") +
"<br>");
  }
}
%>
<a href="login.jsp">Logout</a>
</body></html>
```

//web.xml

```
<web-app>
<servlet>
```

```

        <servlet-name>LoginServlet</servlet-name>
        <servlet-class>servlet.LoginServlet</servlet-class>
    </servlet>
    <servlet-mapping>
        <servlet-name>LoginServlet</servlet-name>
        <url-pattern>/LoginServlet</url-pattern>
    </servlet-mapping>

    <servlet>
        <servlet-name>BookServlet</servlet-name>
        <servlet-class>servlet.BookServlet</servlet-class>
    </servlet>
    <servlet-mapping>
        <servlet-name>BookServlet</servlet-name>
        <url-pattern>/BookServlet</url-pattern>
    </servlet-mapping>

    <servlet>
        <servlet-name>BorrowServlet</servlet-name>
        <servlet-class>servlet.BorrowServlet</servlet-class>
    </servlet>
    <servlet-mapping>
        <servlet-name>BorrowServlet</servlet-name>
        <url-pattern>/BorrowServlet</url-pattern>
    </servlet-mapping>
</web-app>

```

Code Explanation (Module-wise)

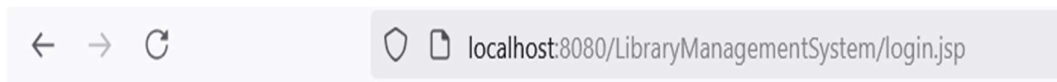
Component	Purpose and Functionality
login.jsp	The login page where users select their role (librarian or user) and submit their login. Redirects to LoginServlet.java.
LoginServlet.java	Validates login. Sets role ("librarian" or "user") in session. Redirects to respective dashboard (librarianDashboard.jsp or userDashboard.jsp).
BookServlet.java	Librarian adds books and stores them in session (books list). Also handles issuing books to users.
BorrowServlet.java	Records borrowing history and stores it in session (borrowHistory). Updates book availability.
Book.java	Model class to define book details (title, author, availability).
Borrow.java	Model class to define borrow records (user, book, date).
search.jsp	Displays available books for users. Uses session role checking to ensure only users access it.
borrowHistory.jsp	Shows user's borrowing history. Accessible only by users after login.

Component	Purpose and Functionality
userDashboard.jsp	User's homepage with navigation links (Search Books, View History, Logout). Checks user role.
librarianDashboard.jsp	Librarian's homepage with navigation links (Add Book, Issue Book, Logout). Checks librarian role.
web.xml	Maps Servlets (LoginServlet, BookServlet, BorrowServlet) and configures session timeout.

Output

1. Login Page (login.jsp)

- Enter username and password.
- Select role: Librarian or User.



2. Librarian Dashboard (librarianDashboard.jsp)

- Add new books.
- Issue books to users.

Librarian Dashboard

Book ID: Title:

Book List

- 1 - Ryu - Borrowed
- 2 - Naruto - Available
- 3 - JJK - Borrowed
- 4 - AOT - Borrowed
- 5 - Tokyo Revengers - Available
- 6 - Demon Slayer: Kimetsu no Yaiba - Available
- 7 - The God of High School - Available
- 8 - Bleach - Available
- 9 - One Piece - Borrowed
- 10 - Jackie Chan Adventures - Available

[Logout](#)

3. User Dashboard (userDashboard.jsp)

- Search for books.
- View borrowing history.

User Dashboard

Search Book:

[View Borrow History](#)

[Logout](#)

4. Book Search Page (search.jsp)

- Users enter search terms to find books.

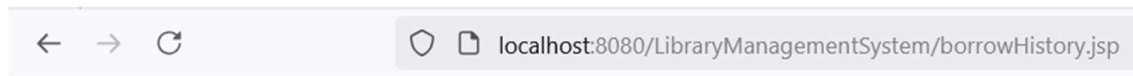
Search Results

7 - The God of High School

[Back](#)

5. Borrowing History Page (borrowHistory.jsp)

- Shows the user's borrowed books dynamically.



Your Borrowing History

Book ID: 4 Date: Fri May 09 23:35:38 IST 2025

Book ID: 1 Date: Fri May 09 23:35:48 IST 2025

Book ID: 3 Date: Fri May 09 23:35:59 IST 2025

Book ID: 9 Date: Fri May 09 23:36:34 IST 2025

[Back](#)

Conclusion

This Library Management System demonstrates role-based access using sessions in Servlets and JSP. It fulfills basic requirements of adding books, issuing them, searching, and tracking borrowing records. All data is maintained in server-side sessions for this version. This project can be extended further by integrating with a database (like MySQL) for persistent storage.